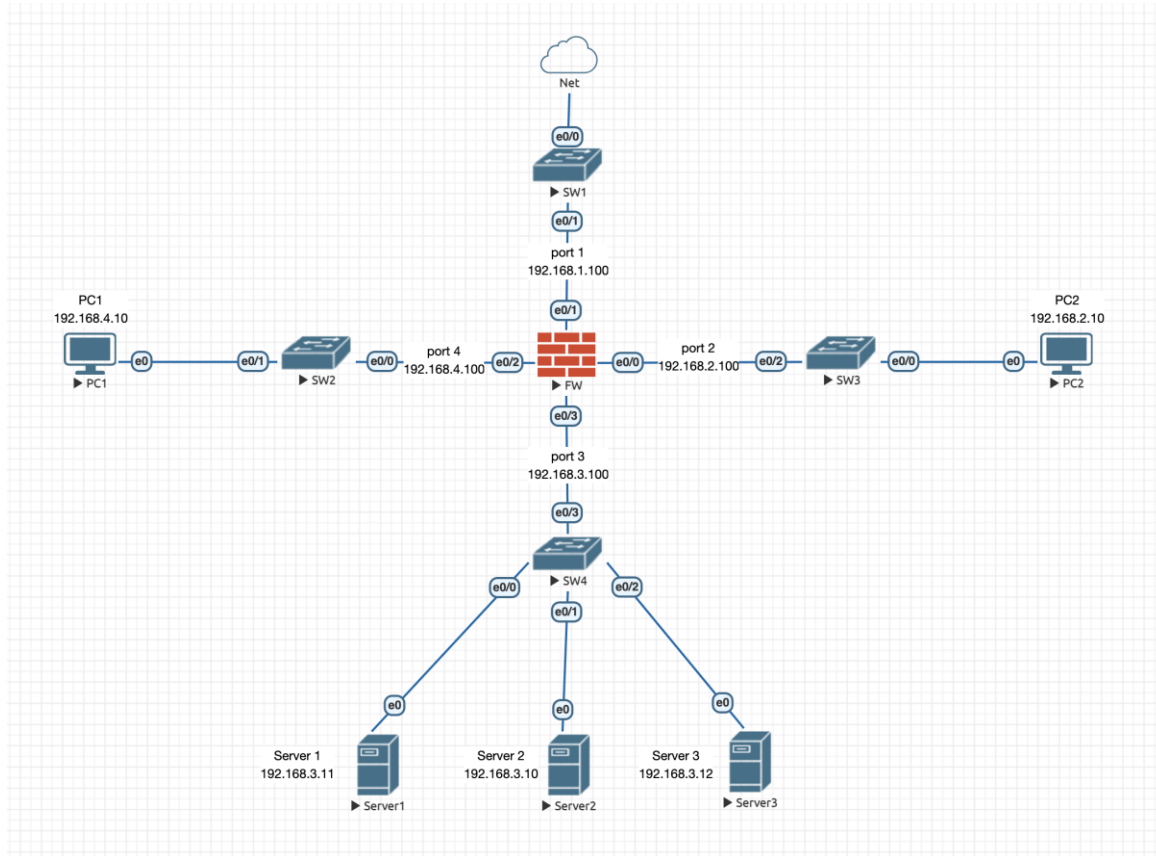


Internetwork Security Project

Topology:



Configured Nodes:

CONFIGURED NODES

ID	NAME	TEMPLATE	BOOT IMAGE	CPU	CPU LIMIT	IDLE PC	NVRAM (KB)	RAM (MB)	ETH	SER	CONSOLE	ICON	STARTUP-CONFIG	ACTIONS
1	SW1	iol	ib6bi_lir	n/a	n/a	n/a	1024	512	1	0	telnet	Switch.png	None	▶ ⚙️ 👤 ⚡ ⌂
2	SW2	iol	ib6bi_lir	n/a	n/a	n/a	1024	512	1	0	telnet	Switch.png	None	▶ ⚙️ 👤 ⚡ ⌂
3	SW3	iol	ib6bi_lir	n/a	n/a	n/a	1024	512	1	0	telnet	Switch.png	None	▶ ⚙️ 👤 ⚡ ⌂
4	PC1	c1710	c1710-b	n/a	n/a	0x80369ac4	128	512	n/a	n/a		PC-2D-Desktop-Generic-S.svg	None	▶ ⚙️ 👤 ⚡ ⌂
5	PC2	c1710	c1710-b	n/a	n/a	0x80369ac4	128	512	n/a	n/a		PC-2D-Desktop-Generic-S.svg	None	▶ ⚙️ 👤 ⚡ ⌂
6	Server1	c1710	c1710-b	n/a	n/a	0x80369ac4	128	512	n/a	n/a		Server.png	None	▶ ⚙️ 👤 ⚡ ⌂
7	Server2	c1710	c1710-b	n/a	n/a	0x80369ac4	128	512	n/a	n/a		Server.png	None	▶ ⚙️ 👤 ⚡ ⌂
8	Server3	c1710	c1710-b	n/a	n/a	0x80369ac4	128	512	n/a	n/a		Server.png	None	▶ ⚙️ 👤 ⚡ ⌂
9	SW4	iol	ib6bi_lir	n/a	n/a	n/a	1024	512	2	0	telnet	Switch.png	None	▶ ⚙️ 👤 ⚡ ⌂
10	FW	iol	ib6bi_lir	n/a	n/a	n/a	1024	1024	1	0	telnet	Firewall-2D-BrickRed-S.svg	Exported	▶ ⚙️ 👤 ⚡ ⌂

- Qemu files do not work on my MacBook due to the macOS Hypervisor Framework and VirtualBox's nested virtualization support on Intel hardware; KVM hardware acceleration is unavailable in my EVE-NG environment. This prevents heavy, closed-source images like FortiOS and Windows 7 from booting. I used i86bi and c1710 to configure the nodes in this topology.

Subnet IP addresses

```

FIREWALL#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              192.168.2.100   YES manual up          up
Ethernet0/1              192.168.1.100   YES manual up          up
Ethernet0/2              192.168.4.100   YES manual up          up
Ethernet0/3              192.168.3.100   YES manual up          up
FIREWALL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 192.168.1.1
     192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C     192.168.1.0/24 is directly connected, Ethernet0/1
L     192.168.1.60/32 is directly connected, Ethernet0/1
L     192.168.1.100/32 is directly connected, Ethernet0/1
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.2.0/24 is directly connected, Ethernet0/0
L     192.168.2.100/32 is directly connected, Ethernet0/0
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.3.0/24 is directly connected, Ethernet0/3
L     192.168.3.100/32 is directly connected, Ethernet0/3
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.4.0/24 is directly connected, Ethernet0/2
L     192.168.4.100/32 is directly connected, Ethernet0/2

```

PC1 pings PC2 & vice versa

```
PC1>ping 192.168.2.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/47/88 ms

PC2#ping 192.168.4.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.4.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/36/52 ms
```

PC1 can access web servers 1 & 2

```
PC1>telnet 192.168.3.10 80
Trying 192.168.3.10, 80 ... Open
```

```
PC1>telnet 192.168.3.11 80
Trying 192.168.3.11, 80 ... Open
```

PC1 fails to access web server 3

```
PC1>telnet 192.168.3.12 80
Trying 192.168.3.12, 80 ...
% Destination unreachable; gateway or host down
```

PC2 can access web server 3

```
PC2#telnet 192.168.3.12 80
Trying 192.168.3.12, 80 ... Open
```

PC2 fails to access web servers 1 & 2

```
PC2#telnet 192.168.3.10 80
Trying 192.168.3.10, 80 ...
% Destination unreachable; gateway or host down

PC2#telnet 192.168.3.11 80
Trying 192.168.3.11, 80 ...
% Destination unreachable; gateway or host down
```

Server3 is accessed through internet on IP address 192.168.1.60

```
FIREWALL>show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
--- 192.168.1.60       192.168.3.12     ---               ---
```

Firewall is connected to gateway 192.168.1.1

```
FIREWALL>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 192.168.1.1
      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Ethernet0/1
L      192.168.1.60/32 is directly connected, Ethernet0/1
L      192.168.1.100/32 is directly connected, Ethernet0/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.2.0/24 is directly connected, Ethernet0/0
L      192.168.2.100/32 is directly connected, Ethernet0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.3.0/24 is directly connected, Ethernet0/3
```

Firewall Rules

```
FIREWALL#show access-lists
Standard IP access list 1
 10 permit 192.168.2.0, wildcard bits 0.0.0.255
 20 permit 192.168.3.0, wildcard bits 0.0.0.255
 30 permit 192.168.4.0, wildcard bits 0.0.0.255
Standard IP access list 10
 10 permit 192.168.2.0, wildcard bits 0.0.0.255
 20 permit 192.168.3.0, wildcard bits 0.0.0.255
 30 permit 192.168.4.0, wildcard bits 0.0.0.255
Extended IP access list CISCO-CWA-URL-REDIRECT-ACL
100 deny udp any any eq domain
101 deny tcp any any eq domain
102 deny udp any any eq bootps any
103 deny udp any any eq bootpc
104 deny udp any eq bootpc any
105 permit tcp any any eq www
Extended IP access list FIREWALL_ACL
 10 permit icmp 192.168.4.0 0.0.0.255 192.168.2.0 0.0.0.255 (10 matches)
 20 permit icmp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255 (10 matches)
 30 permit tcp 192.168.4.0 0.0.0.255 host 192.168.3.10 eq www (14 matches)
 40 permit tcp 192.168.4.0 0.0.0.255 host 192.168.3.11 eq www (14 matches)
 50 deny tcp 192.168.4.0 0.0.0.255 host 192.168.3.12 eq www (2 matches)
 60 permit tcp 192.168.2.0 0.0.0.255 host 192.168.3.12 eq www (14 matches)
 70 deny tcp 192.168.2.0 0.0.0.255 host 192.168.3.10 eq www (2 matches)
 80 deny tcp 192.168.2.0 0.0.0.255 host 192.168.3.11 eq www (2 matches)
 90 permit ip 192.168.2.0 0.0.0.255 any (44 matches)
100 permit ip 192.168.3.0 0.0.0.255 any (51 matches)
110 permit ip 192.168.4.0 0.0.0.255 any (52 matches)
130 permit ip 192.168.1.0 0.0.0.255 any (527 matches)
150 permit icmp any 192.168.2.0 0.0.0.255 echo-reply
160 permit icmp any 192.168.3.0 0.0.0.255 echo-reply
170 permit icmp any 192.168.4.0 0.0.0.255 echo-reply
190 permit icmp any any echo-reply
200 permit icmp 192.168.1.0 0.0.0.255 any
210 deny ip any any log (2 matches)
Extended IP access list INTERNET_IN
 5 permit icmp any any echo-reply
 6 permit icmp any any
10 permit tcp any host 192.168.1.60 eq www
20 permit tcp any any established
30 deny ip any any log (176 matches)
```

These rules define:

PC1 <->PC2:

permit icmp 192.168.4.0 0.0.0.255 192.168.2.0 0.0.0.255

permit icmp 192.168.2.0 0.0.0.255 192.168.4.0 0.0.0.255

PC1 -> Server1 & Server2 ONLY:

```
permit tcp 192.168.4.0 0.0.0.255 host 192.168.3.10 eq 80
permit tcp 192.168.4.0 0.0.0.255 host 192.168.3.11 eq 80
deny tcp 192.168.4.0 0.0.0.255 host 192.168.3.12 eq 80
```

PC2 -> Server3 ONLY:

```
permit tcp 192.168.2.0 0.0.0.255 host 192.168.3.12 eq 80
deny tcp 192.168.2.0 0.0.0.255 host 192.168.3.10 eq 80
deny tcp 192.168.2.0 0.0.0.255 host 192.168.3.11 eq 80
```

Allow all subnets to access the Internet:

```
permit ip 192.168.2.0 0.0.0.255 any
permit ip 192.168.3.0 0.0.0.255 any
permit ip 192.168.4.0 0.0.0.255 any
```

Allow replies from the internet to all the subnets:

```
permit ip 192.168.2.0 0.0.0.255 any
permit ip 192.168.3.0 0.0.0.255 any
permit ip 192.168.4.0 0.0.0.255 any
permit ip 192.168.1.0 0.0.0.255 any
```

Allow icmp replies to allow pings to the internet:

```
permit icmp any 192.168.2.0 0.0.0.255 echo-reply
permit icmp any 192.168.3.0 0.0.0.255 echo-reply
permit icmp any 192.168.4.0 0.0.0.255 echo-reply
```

```
permit icmp any any echo-reply
```

```
permit icmp 192.168.1.0 0.0.0.255 any
```

Allow Internet access inside the subnets:

```
permit icmp any any echo-reply
```

```
permit icmp any any
```

```
permit tcp any host 192.168.1.60 eq www
```

```
permit tcp any any established
```

Devices accessing the Internet

- Although NAT, ACLs, and routing were correctly configured and internally verified, Internet access for internal devices could not be achieved due to EVE-NG Cloud bridging and IOS firewall limitations, not configuration errors.

```
PC1>ping 8.8.8.8
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
```

```
.....
```

```
Success rate is 0 percent (0/5)
```

```
Success rate is 0 percent (0/5)
```

```
PC2>ping 8.8.8.8
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
```

```
.....
```

```
Success rate is 0 percent (0/5)
```

```
SERVER1>ping 8.8.8.8
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
```

```
.....
```

```
Success rate is 0 percent (0/5)
```

```
SERVER2>ping 8.8.8.8
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
```

```
.....
```

```
Success rate is 0 percent (0/5)
```